MPhil Thesis Structure

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Contents

1 Introduction

- Discuss pharmaceutical innovation
- Contributions in this paper - state the general problem / area of research.
- Part of the contribution is to collect and collate this data which is relevant for the question.
- Summary of Medicaid
- Discuss ACA: What changed in Medicaid, and what this means
- How is the ACA related to pharmaceutical innovation
- Why is it relevant?
- How will this paper be structured?

2 Literature Review

- Literature on Medicaid / ACA - show that we're looking into literature that is relatively active. If they're new papers that works better. Want to give a feeling that when writing the paper we're speaking to an audience that is relatively large, and the audience looking at this policy is relatively large for different types of outcomes but this is just to show that the work is nested in an active research field.
- Public procurement and pharmaceutical pricing
- Market size and pharmaceutical innovation
- Dynamic innovation incentives

3 The ACA

- PDL
- Rebates
- Market size
- Insurance
- Flag that we're not using staggered roll out to manage expectations

4 Data

- SDUD
- Drugs@FDA:
- Patents
- PDL
- RxNorm

5 Methodology

- RQ1: Overall innovation response to market size. FEOLS, FE Poisson, Ex-ante, Ex-post
- RQ2: Disinentangling the effects of market sizing and pricing pressure. FEOLS, FE Poisson, Ex-ante, Ex-post
- RQ3: Innovation portfolio response - the DDD estimator.
- Calculation of Medicaid Share
- Calculation of PDL exposure
- Parallel trends tests
- Firm level variation
- PDL Counterfactual analysis - potential research question?

6 Results

- Parallel trends
- RQ1
- RQ2
- RQ3

7 Theoretical Model

- Description of game
- Stage 1
- Stage 2
- Stage 3
- State dependent innovation - period 1 outcomes have implications for period 2 effort.
- Mapping theoretical states to empirical outcomes
- Equilibrium Analysis
- Comparative Statics
- Explaining the results

8 Policy Implications

How do you weigh the present benefits of consumer surplus with the cost to innovation in the future. As a social planner, planning for your rebate you need to account for this. Discussions should mostly look into this

- Tension between states wanting lower net drug prices while firms need sufficiently high expected returns to justify R&D spending
- Theoretical model highlights rebate mechanism is largely responsible for the current results
- Lower harm inflicted to firms through rebate mechanism
- Lower the cost of R&D through subsidies or other incentives

9 Extensions

- Currently map data to ATC, more granularity would be better.
- Clinical trial data
- PDL data from other states
- Theoretical model is illustrative, include an N player game with multiple winners
- Demand is assumed to be perfectly inelastic
- Logit model

10 Conclusion

Just a conclusion